Understanding cache block

Sparsh Mittal

Example 1

- A cache has 4MB cache capacity.
- Find the number of blocks in it, if block size is (a) 64B and (b) 128B
- Answer: (a) 4MB/64B = 65,536
- (b) 4MB/128B = 32,768

- ➔ On increasing the block size, we can keep less number of blocks in the cache

We will now discuss an example of a small cache.
We will change the block size and show the cache operation

Case-1: Block size is 1B (block offset has 0 bits)

- If I access address 7, which address I get? Answer: 7
- If I access address 4, which address I get? Answer: 4
- If I first access address 7 and then 4, will access to 4 lead to a hit?
- Answer: NO.
- If I access address 14, which addr I get?
- Answer: 14

Case-1: Block size is 2B (block-offset has 1 bit)

- If I access address 7, which addresses I get? Answer: 6 and 7

Reason: 6=**011**0, 7=**011**1

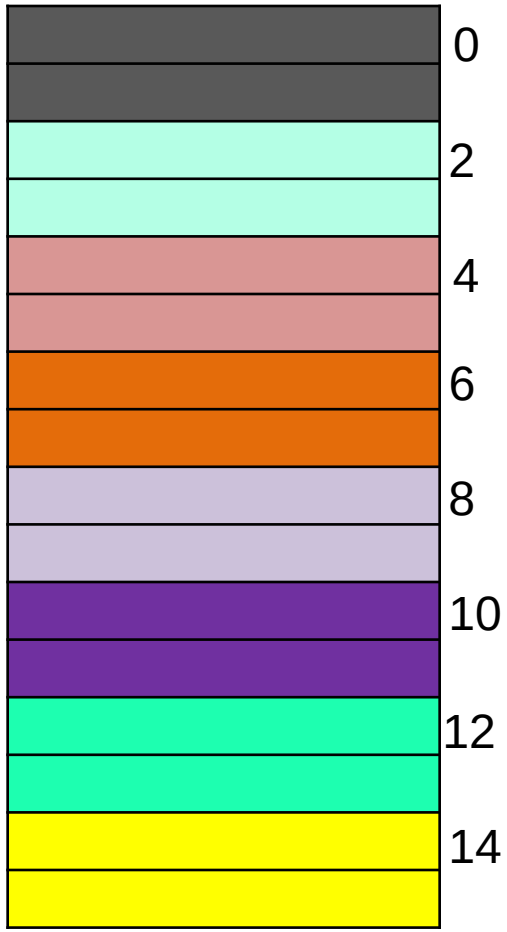
On ignoring offset bits, remaining bits are same. Both 6 and 7 are part of the same block

- If I access address 14, then?

Answer: 14 and 15

Reason: 14=**111**0, 15=**111**1

Both 14 and 15 are in same block



Why these bits are called block-offset bits?

- Assume block-offset has only 1 bit (i.e., block-size is 2B)
- If I access address 7, I will get 6 and 7

6=**011**<u>0</u>, 7=**011**<u>1</u>

- Now, within that 2B block,
  - The offset of address 6 is 0
  - The offset of address 7 is 1
- Hence, we call these bits as block-offset bits.

Case-1: Block size is 4B (block-offset has 2 bits)

- If I access address 7, which addresses I get?
- Answer: 4, 5, 6 and 7

4=**01**00, 5=**01**01 6=**01**10, 7=**01**11

On ignoring offset bits, remaining bits are same.

4, 5, 6 and 7 are part of the same block

- If instead, we had accessed 5, which addressed I would have got?
- Answer: 4, 5, 6, 7

Case-1: Block size is 4B (block-offset has 2 bits)

- If I first access address 7 and then 4, will access to 4 lead to a hit?
- Answer: YES.
- If I access address 14, which addr I get?
- Answer: 12, 13, 14 and 15
- 12=11<u>00</u>, 13=11<u>01</u>, 14=11<u>10</u>, 15=11<u>11</u>
- 12,13, 14 and 15 are in same block

Tradeoff in determining block-size

- Larger blocks should reduce miss rate
  - Due to spatial locality
- But in a fixed-sized cache
- Larger blocks => fewer of them => pollution
- More competition => increased miss rate
- Larger miss penalty
- Can override benefit of reduced miss rate

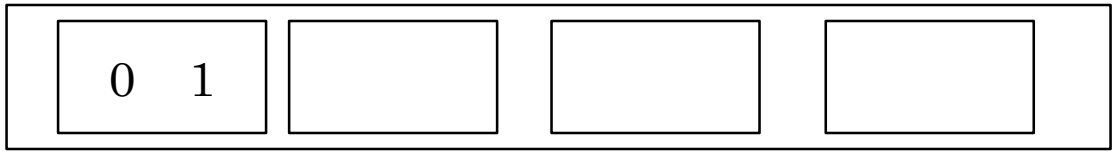
How large cache blocks reduce #misses

- Consider an 8B fully-associative cache.
- Memory addresses are 0 to 15
- We access 0, 1, 2, 3, 4, ...15 sequentially
- **Case 1:** cache block size is 1B.
- We make 16 accesses, all of them are misses.
- **Case 2:** cache block size is 2B.
- We make 16 accesses, 8 of them are misses.
- **Case 3:** cache block size is 4B.
- We make 16 accesses, 4 of them are misses.

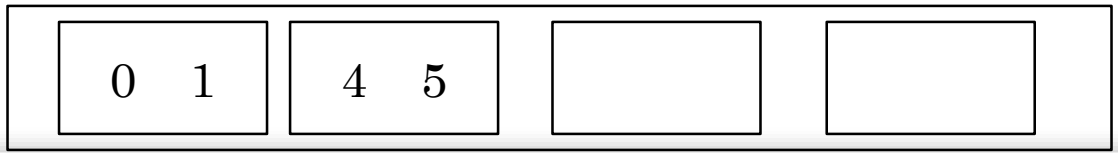
How large blocks can lead to pollution [1 of 3]

- Consider an 8B fully-associative cache.
- Memory addresses are 0 to 15
- We access 0, 4, 8, 12, 1, 5, 9, 13, 2, 6, 10, 14, 3, 7, 11, 15
- Case 1: 1B cache block. 16 accesses, all misses
- Case 2: 2B cache block. 16 accesses

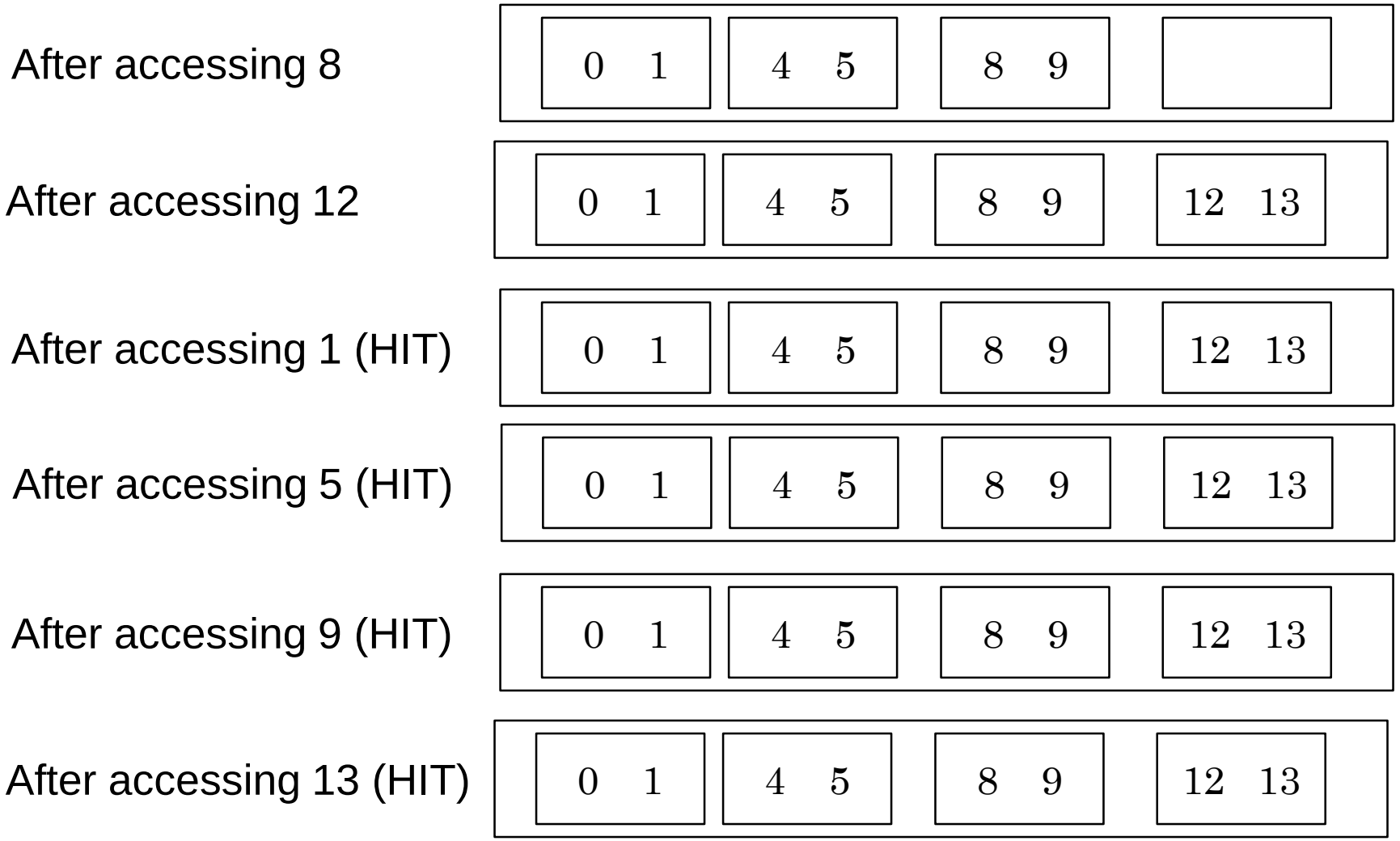
After accessing 0



After accessing 4



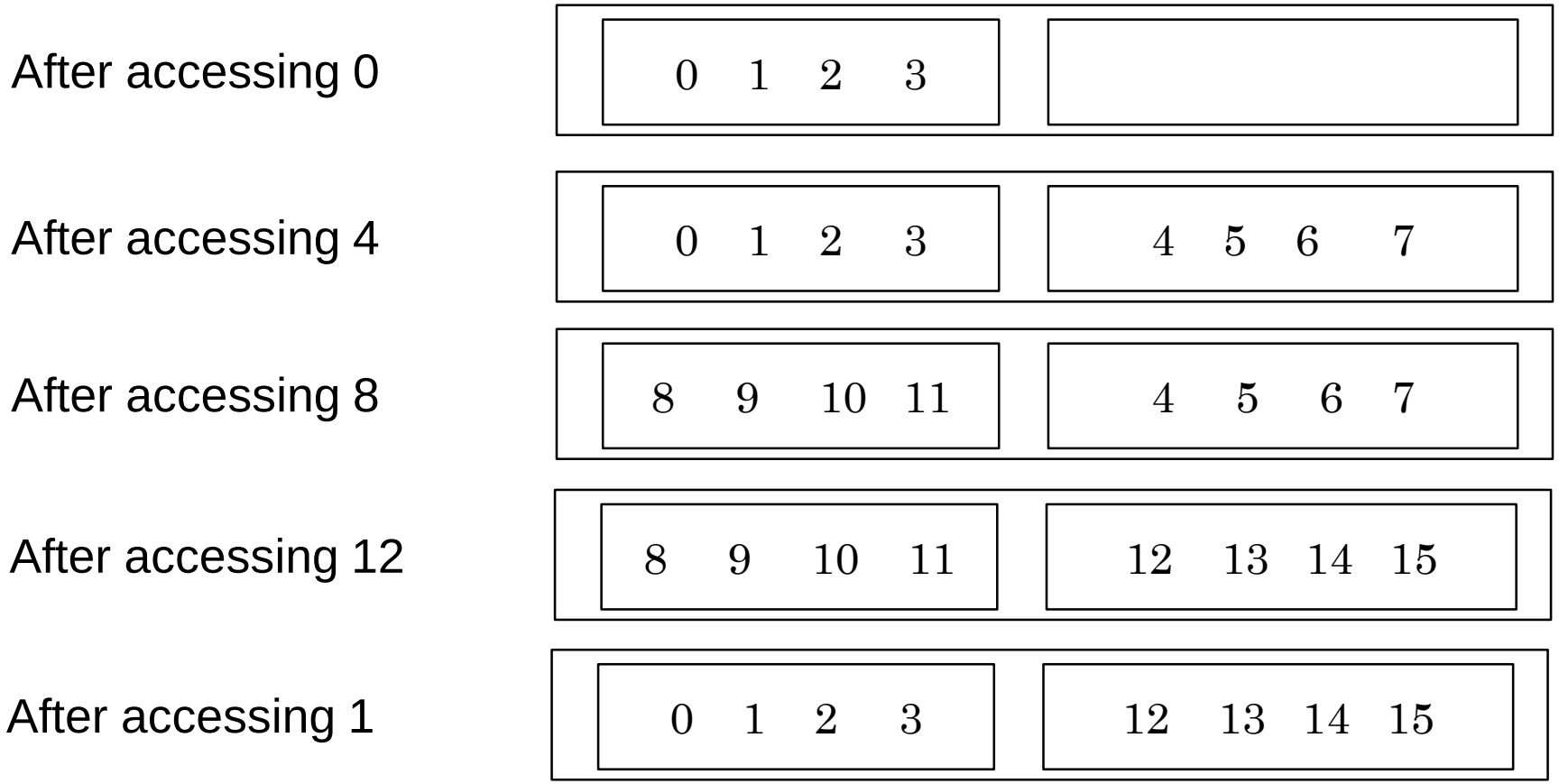
How large blocks can lead to pollution [2 of 3]



Continuing like this, out of 16 accesses, 8 will hit.

How large blocks can lead to pollution [3 of 3]

- Case 3: 4B cache block. 16 accesses



Continuing like this, all the 16 accesses will miss